Can the app find strong players' positions like his, whose move works in his?

Matches by pawns, pieces and kings from 359,845 strong players' games, against random matches with the same pieces, refereed by Stockfish

Chess Coach research lab

Abstract

Can the app find positions from strong players' games so like a position from his games that the strong player's move works in his too? For 4,807 quiet middlegame and endgame positions from his 647 10-minute games, an index of 15,792,886 positions from 359,845 Lichess games between players rated 2200 or more gives the three nearest positions with the same pieces, by a distance over the pawns, the pieces and the kings, and three random positions with the same pieces, and Stockfish says which of the strong players' moves are sound in his position. By the rule fixed before any move was scored, likeness works: the nearest positions' moves are sound 26.8% and 27.3% of the time on the two halves of his games, against 20.8% and 21.6% for random ones, a lift of 6.1 and 5.8 points with both intervals above 0, and both controls pass. It works too weakly to give advice: his own move is sound 52.0% and 55.3% of the time, and Maia-2's most likely move at his band 60.7% and 64.9%. A match's move is sound more often the nearer its position, from 22% and 23% at a distance of 10 or more to 74% and 75% within 2, but most of his positions have no match nearer than 10, and the few with one within 2 do no better there than Maia-2's move. Added after the run: 16 times as many games moves the median distance of his nearest match only from 19 to 14. The app takes no search for similar positions from this. The lift is largest where the app names the pawn structure, so what carries from a strong player's game to his is more likely the plan the structure calls for than the next move.

1 The question

A coach explains a middlegame or endgame decision with an example: this is like a game where a strong player had the same pawns and pieces and played this move. The example teaches a plan only when the positions are alike in the way that matters, so that the strong player's move works in the student's position too. The app keys pawn skeletons exactly and has no search for similar positions. This study asks whether one would find such examples for him. For a quiet middlegame or endgame position from his games, it finds the positions from games between strong players nearest to his by a distance over the pawns, the pieces and the kings, and asks whether the strong players' moves there are sound in his position more often than the moves of strong players from random positions with the same pieces.

If likeness works, a "this is like…" example in the review or a drill can show a strong player's position like his with the move played there, and the likeness is part of why the move works. If a random position with the same pieces does as well, the likeness an example claims is not why its move works, and the app has no reason to search for one.

2 How it was measured

His positions. His 647 10-minute games in the app's archive at commit `f4bd63f19b`, in order of their end time: the first 324 (July 22 to August 23) and the last 323 (August 23 to October 15) are the two halves. Nothing is fitted, so the halves are two replications. Each of his moves enters when the position is past the opening by the app's divider, he is not in check and has at least two legal moves, the board holds eight men or more (fewer is the tablebase's ground), his win% before the move is between 10 and 90 by the archive's evaluation, and his clock shows at least 30 seconds. That leaves 8,896 positions, 4,492 in the first half and 4,404 in the second; the referee below keeps the quiet ones.

Strong players' positions. Lichess's September 2026 database (CC0) gives the first 400,000 rated blitz, rapid and classical games in file order between two players rated 2200 or more that ended normally or on time and have at least 30 plies, out of 23,231,008 games read. Most are blitz: 365,813 blitz, 33,040 rapid and 1,147 classical. One game in ten, by a seeded hash of its id, is held out (40,155 games); the other 359,845 are the index. Every position before a move in the index games that is past the opening by the app's divider, not in check, holds eight men or more, and where the mover's clock shows at least 30 seconds enters the index with the move played there: 15,792,886 positions.

Alike. Every position is read from the side to move: when Black is to move, the board is mirrored top to bottom and the colours swapped, so the mover always plays up the board, and moves are mirrored the same way. Two positions can match only when each side has the same number of queens, rooks, knights and pawns, and of bishops on each colour of square. The *distance* between two such positions adds three parts:

- **pawns**: the squares where one position has a pawn of a side and the other does not, so a pawn one square further on counts 2;
- **pieces**: for each side and each kind of piece (knight, bishop on each colour, rook, queen), the two positions' pieces paired to make the total least, each pair counting the king-step distance between its squares, at most 2;
- **kings**: each king's king-step distance between the two positions, at most 3.

The weights were set without looking at any position: the pawns, which fix the plans, count the most, and a piece counts at most a moved pawn. A match *transfers* when its move, read in his position, is legal there and moves the same kind of piece from the same square.

The moves compared. In each position, **T** is the strong players' moves from the three nearest index positions with the same pieces, from three different games, whose moves transfer, ties in distance broken by a seeded hash. **R** is the moves from three index positions with the same pieces drawn at random, from three more games, none of them T's, whose moves transfer, after at most 20,000 draws. **R2** is a second random draw the same way, from games none of T's or R's. His positions also have **N**, Maia-2's most likely move at his band; **U**, its most likely move at the band 200 points above; and **A**, his move. His band is his rating at the game with the app's offset of 250 points, the band above the same of his rating plus 200, each conditioned on the opponent's band. A position is *covered* when T, R and R2 each find three moves.

The referee. Stockfish through the app's engine service at the grading budget of 250,000 nodes, with its hash cleared before each position and no tablebase, gives the best move and scores every move compared. A move is *sound* when it gives up no more than 3 win% against the best, by the app's own test for an alternative move. A position is *quiet* when the best move is not a capture, a check or a promotion and the win% it gives the side to move is between 10 and 90; only quiet positions are scored. A score that is not exact counts as missing, and a position with a missing score leaves the analysis.

The measures. In each position, an arm's *soundness* is the share of its three moves that are sound. The *lift* is T's soundness minus R's, averaged over positions, paired by position. 2,000 bootstrap draws resample his games within each half, or the strong players' positions; intervals are the 95% percentile intervals.

The rule, fixed before any move was scored. Two controls come first, and if either fails the study gives no verdict.

1. **The planted twin.** 1,200 held-out strong players' positions, one a game, covered and quiet by the same filters (a game tries another of its positions, at most three, when one is not). Planted in the index with its own move, each would be its own nearest match, at distance 0, with a move that transfers. The twin's move must be sound more often than R's moves: the interval of the twin's soundness minus R's lies above 0.
2. **The null.** R2 against R, in place of T against R, must read neither as likeness working nor as likeness misleading.

Then, on his positions, *likeness works* when the lift's interval lies above 0 on both halves, *likeness misleads* when it lies below 0 on both, and it is *no better than a random match* otherwise. A half with fewer than 300 scored positions is *not enough data*, whatever the numbers say. The rule and the scripts went to the lab before any move was scored; the coverage of the final index was counted after the rule and before any move was scored.

3 Coverage

				Distance of the nearest match			
	Positions	Covered (%)	Bucket (median)	10%	25%	Median	75%
His, first half	4,492	77.9	1,408	8	11	15	18
His, second half	4,404	84.0	2,348	8	11	14	18
Strong players, held out	3,000	89.7	–	6	10	15	18

Table 1: How much of each set of positions the index covers, counted before any move was scored: the share of positions for which T, R and R2 (T and R for the strong players' positions) each find three matches whose moves transfer, the number of index positions with the same pieces, and the distance of the nearest match that transfers. The strong players' positions are a sample of the held-out ones.

The index covers most of his positions but rarely closely (Table 1). It finds three matches for each arm in 78% of his candidate positions in the first half and 84% in the second, and in 90% of a

sample of 3,000 held-out strong players' positions. The nearest match that transfers has a median distance of 15 and 14 in the two halves, and 15 for the strong players' positions: a typical match differs from his position in several pawns and most of the pieces.

Of the covered positions, the referee keeps the quiet ones: 2,309 positions from 280 games in the first half and 2,498 from 287 games in the second. In 1,159 and 1,197 positions the best move is a capture, a check or a promotion, and in 33 and 6 the referee reads the game as decided. Every score was exact.

4 The controls

The planted twin passes. In 1,200 held-out strong players' positions, the strong player's own move is sound 66.6% of the time (63.9 to 69.3) and a random match's move 20.1% (18.5 to 21.7); the difference, 46.5 points (43.6 to 49.2), lies far above 0. The referee tells a move that belongs to a position from one that only fits it. A strong player's own move is sound in only about two thirds of their positions: most of the index is blitz, and a strong player's blitz move is often more than 3 win% from Stockfish's best. That is no ceiling for a match: the nearest matches of strong players' positions are sound 88% of the time (Table 3), since a move that players chose in nearly the same position is sound more often than one player's move in one game.

The null passes. R2 against R reads 0.3 points (−0.8 to 1.5) in the first half and −0.5 points (−1.5 to 0.6) in the second; both intervals hold 0.

5 The verdict: likeness works

Move	First half	Second half
Positions (games)	2,309 (280)	2,498 (287)
T, the matches	26.8 (25.0, 28.7)	27.3 (25.3, 29.3)
R, random matches	20.8 (19.1, 22.4)	21.6 (19.8, 23.5)
R2, a second draw	21.1 (19.4, 22.9)	21.1 (19.2, 23.0)
N, his band	60.7 (58.3, 63.1)	64.9 (62.2, 67.6)
U, the band above	62.4 (60.0, 64.8)	65.6 (62.9, 68.4)
A, his move	52.0 (49.7, 54.1)	55.3 (52.8, 58.0)
Lift, T minus R (points)	6.1 (4.6, 7.5)	5.8 (4.4, 7.2)
Null, R2 minus R (points)	0.3 (−0.8, 1.5)	−0.5 (−1.5, 0.6)

Table 2: The share of each arm's moves that are sound in his positions, in %, with its 95% interval, and the lift and the null in points. A move is sound when it gives up no more than 3 win% against Stockfish's best.

The strong players' moves from the three nearest positions are sound in his position 26.8% and 27.3% of the time on the two halves, and the random matches' 20.8% and 21.6% (Table 2). The lift is 6.1 points (4.6 to 7.5) and 5.8 points (4.4 to 7.2), above 0 on both halves, so by the rule likeness works. A second random draw does no better than the first: R2's moves are sound 21.1% and 21.1% of the time.

It works far too weakly to give advice. His own move is sound 52.0% and 55.3% of the time, Maia-2's most likely move at his band 60.7% and 64.9%, and at the band above 62.4% and 65.6%: a strong player's move from a position like his is sound about half as often as his own. On average the matches' moves give up 15.5 and 15.3 win% against the best, about one point less than the random matches' (16.5 and 16.4), while his own moves give up 6.1 and 5.7 (Table 5). The matches' moves are the referee's best 7.4% and 7.0% of the time, the random matches' 4.2% and 3.2%.

6 What the positions describe

	His, first half			His, second half			Strong players, held out		
Distance	Matches	Sound (%)	From T	Matches	Sound (%)	From T	Matches	Sound (%)	From T
0 to 2	38	73.7 (50.0, 89.5)	100	44	75.0 (55.6, 90.2)	100	122	87.7 (81.0, 93.3)	100
3 to 5	158	57.6 (47.1, 66.9)	100	177	53.1 (43.6, 62.3)	100	173	59.0 (50.0, 68.5)	100
6 to 9	733	43.9 (39.3, 48.6)	100	787	44.2 (39.2, 49.3)	99	427	45.9 (40.5, 51.6)	99
10 or more	12,925	22.1 (20.5, 23.7)	46	13,980	22.8 (21.2, 24.6)	46	6,478	21.2 (19.7, 22.7)	44

Table 3: Every T and R match by its own distance from the position: how many there are, the share whose move is sound, and the share of them, in %, that come from T. Intervals resample the games.

Nearer is better. A match's move is sound more often the nearer its position (Table 3): 74% and 75% of the matches within 2 on the two halves, 58% and 53% from 3 to 5, 44% and 44% from 6 to 9, and 22% and 23% at 10 or more. Strong players' own positions show the same, from 88% within 2 to 21% at 10 or more. Almost every match within 9 is one of T's; at 10 or more, where all of R's are, a little under half are T's.

Half	Positions of	n	T (%)	R (%)	Lift (points)
First	Endgame	621	29.0 (24.9, 33.5)	26.1 (22.3, 29.6)	3.0 (0.4, 5.6)
First	Middlegame	1,688	26.0 (24.1, 28.0)	18.8 (17.0, 20.6)	7.2 (5.4, 9.0)
First	A structure the app names	645	31.1 (27.7, 34.5)	21.0 (18.4, 23.8)	10.1 (7.1, 12.9)
First	No named structure	1,664	25.2 (23.0, 27.3)	20.7 (18.5, 22.8)	4.5 (2.9, 6.2)
Second	Endgame	648	29.7 (25.7, 34.0)	27.1 (23.1, 31.7)	2.6 (−0.3, 5.5)
Second	Middlegame	1,850	26.5 (24.6, 28.5)	19.6 (17.9, 21.4)	6.9 (5.4, 8.5)
Second	A structure the app names	706	30.7 (27.5, 34.3)	22.0 (19.1, 25.0)	8.8 (6.5, 11.3)
Second	No named structure	1,792	26.0 (23.7, 28.3)	21.4 (19.3, 23.4)	4.6 (2.9, 6.3)
Strong	Endgame	322	33.9 (29.9, 37.6)	28.3 (24.2, 32.0)	5.6 (2.4, 8.8)
Strong	Middlegame	878	27.6 (25.4, 30.0)	17.0 (15.3, 18.9)	10.6 (8.4, 12.8)

Table 4: T's and R's soundness and the lift by the divider's phase of the position and by whether the app names its pawn structure, with 95% intervals.

Middlegame and structure. The lift is larger in the middlegame, 7.2 and 6.9 points, than in the endgame, 3.0 and 2.6 points, where the second half's interval holds 0 (Table 4). It is larger where the app names the position's pawn structure, 10.1 and 8.8 points, than where it names none, 4.5 and 4.6. In the strong players' own positions the lift is 9.2 points (7.3 to 11.1).

	His, first half	His, second half	Strong players
Win% given up, T	15.53 (14.72, 16.32)	15.33 (14.54, 16.14)	15.15 (14.39, 15.92)
Win% given up, R	16.48 (15.67, 17.30)	16.38 (15.55, 17.18)	17.14 (16.43, 17.88)
Win% given up, R2	16.53 (15.71, 17.35)	16.50 (15.70, 17.32)	–
Win% given up, his move	6.07 (5.66, 6.49)	5.74 (5.33, 6.19)	–
The best move, T (%)	7.4 (6.6, 8.3)	7.0 (6.3, 7.8)	9.6 (8.5, 10.7)
The best move, R (%)	4.2 (3.7, 4.8)	3.2 (2.7, 3.6)	4.2 (3.5, 4.9)
New to him, T (%)	29.4 (27.1, 31.9)	29.9 (27.3, 32.5)	–
New to him, R (%)	28.9 (26.4, 31.3)	29.9 (27.5, 32.4)	–
His move unsound: positions	1,108	1,116	–
some T move sound (%)	34.9 (31.5, 38.2)	36.9 (33.6, 40.4)	–
some R move sound (%)	28.6 (26.0, 31.4)	27.2 (24.8, 29.8)	–

Table 5: The mean win% each arm's moves give up against Stockfish's best (a move that scores above it gives up none); the share of each arm's moves that are the best; the share of positions where a move of the arm is sound, is not his move and has under 10% at his band by Maia-2; and, where his own move is not sound, the share of positions where at least one of the arm's moves is. 95% intervals.

New to him, and his mistakes. A sound move he rarely finds, one that is not his move and that Maia-2 gives under 10% at his band, comes from T in 29.4% and 29.9% of his positions and from R in 28.9% and 29.9%: likeness adds no idea he would miss. Where his own move is not sound, in 1,108 and 1,116 positions, at least one of T's moves is sound in 34.9% and 36.9% of them, and one of R's in 28.6% and 27.2%.

7 Added after the run

These descriptions were added once the rule had been applied; none was named before the run.

Nearest match	Positions	Share (%)	Its move	R	His move	N, his band
0 to 2	47	1.0	78.7 (65.9, 89.4)	38.3 (27.7, 48.1)	72.3 (58.0, 84.6)	83.0 (72.7, 92.5)
3 to 5	156	3.2	56.4 (48.4, 64.6)	28.4 (23.2, 34.0)	63.5 (55.7, 70.8)	80.1 (73.7, 86.4)
6 to 9	639	13.3	43.2 (38.8, 47.7)	27.9 (24.8, 31.1)	61.7 (57.1, 66.1)	68.9 (65.2, 72.6)
10 or more	3,965	82.5	23.9 (22.3, 25.5)	19.6 (18.4, 20.9)	51.9 (50.0, 53.7)	61.0 (59.0, 63.0)

Table 6: Added after the run. His positions, both halves, by the distance of their nearest match: how many, and the share of moves that are sound, in %, for the nearest match's move, R's, his own and Maia-2's most likely move at his band, with 95% intervals resampling his games.

Only a near match competes. By the distance of each position's nearest match (Table 6), 82% of his positions have none nearer than 10, and there the nearest match's move is sound 23.9% of the time, against 51.9% for his own move and 61.0% for Maia-2's most likely move at his band. Only 1.0% of his positions, 47 in all, have a match within 2. There its move is sound 78.7% of the time, about as often as his own move (72.3%) and no more often than Maia-2's (83.0%). These are easier positions: his own move and Maia-2's are sound more often there too.

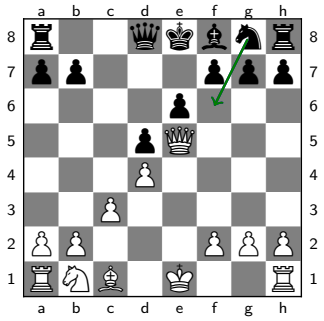
Stream games	Index games	No match	Median distance	Within 2 (%)	Within 5 (%)	Within 9 (%)
25,000	21,768	330	19	0.1	1.2	6.8
50,000	43,625	99	17	0.3	1.8	8.9
100,000	87,416	19	16	0.5	2.3	11.0
200,000	174,760	0	15	0.6	3.3	14.0
400,000	349,647	0	14	1.0	4.2	17.5

Table 7: Added after the run. The nearest match that transfers for each of his scored positions, with the index cut to the positions of the first games of the stream: how many have no match at all, the median distance, a position with no match counting as farther than any, and the share of positions with one within 2, 5 and 9. Index games counts the index games with any of his positions' sets of pieces. With the whole index, every position has its nearest match at the distance T found.

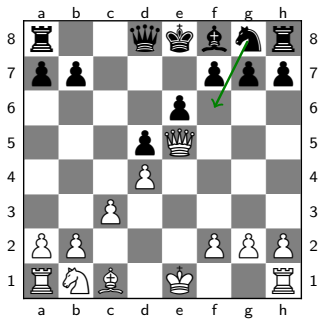
A bigger index helps slowly. Restricting the index to the first games of the stream (Table 7), 16 times as many games moves the median distance of his positions' nearest match from 19 to 14, about one for each doubling, and the share with a match within 2 from 0.1% to 1.0%. Near matches grow with the index, but slowly.

8 Two positions

Two of his positions show what the matches look like (Figure 1): the one with the nearest match of all, and one whose nearest match is at the median distance. They illustrate; they test nothing.

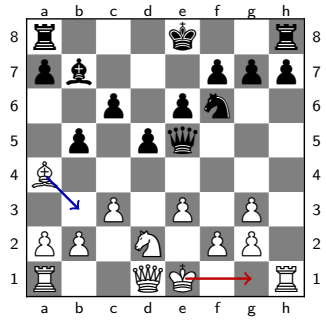


FEN `r2qkbnr/pp3ppp/4p3/3pQ3/3P4/2P5/PP3PPP/RNB1K2R b KQkq - 0 9`

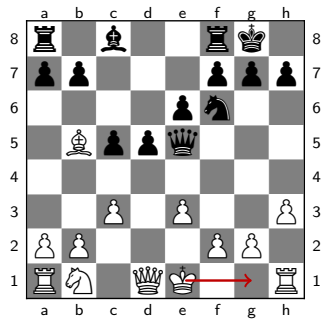


FEN `r2qkbnr/pp3ppp/4p3/3pQ3/3P4/2P5/PP3PPP/RNB1K2R b KQkq - 0 9`

(a) Above, his position: he is Black, in the middlegame, with 52% to win by the best move 9. . . ♘f6. Below, the nearest strong players' position, at distance 0 (pawns 0, pieces 0, kings 0), where 9. . . ♘f6 was played. In his position that move is 9. . . ♘f6, sound at 52%; he played 9. . . ♘f6 (blue), at 52%. Sound here: one of T's three moves and three of R's.



FEN `r3k2r/pb3ppp/2p1pn2/1p1pq3/B7/2P1P1P1/PP1N1PP1/R2QK2R w KQkq - 0 14`



FEN `r1b2rk1/pp3ppp/4pn2/1Bppq3/8/2P1P2P/PP3PP1/RN1QK2R w KQ - 0 11`

(b) Above, his position: he is White, in the middlegame, with 39% to win by the best move 14. ♗c2. Below, the nearest strong players' position, at distance 14 (pawns 6, pieces 6, kings 2), where 11. O-O was played. In his position that move is 14. O-O, not sound at 19%; he played 14. ♗b3 (blue), at 36%. Sound here: zero of T's three moves and zero of R's.

Figure 1: Two of his second-half positions whose nearest match has the same side to move: left, the one whose nearest match is nearest of all; right, the first whose nearest match is at the median distance, 14. The strong player's move is green where it is sound in his position and red where it is not. Stockfish at 250,000 nodes, rerun for this figure.

9 What the app can take

- **No search for similar positions.** Likeness works by the rule, but a strong player's move from a position like his is sound about half as often as his own, and the near matches, the only ones that do as well as Maia-2's most likely move at his band, are rare. A "this is like…" example built this way would mostly show a move that does not work in his position.
- **An example is checked first.** If the app ever shows a strong player's position as an example, the engine checks its move in his position before it is shown, and only a near match is worth showing.
- **A plan, not a move.** The lift is largest where the app names the pawn structure. What carries from a strong player's game to his is more likely the plan the structure calls for than the next move. A next round can match on the structure and test a plan over the next few moves, with its rule fixed first.

10 Caveats

- **A move stands for a plan.** The strong player's next move is tested, not the moves that follow it.
- **The board alone.** The study's plan also named Maia-2's reading of the position as part of alike. The app's Maia-2 gives move probabilities and no reading of who is better, so this round matches on the board alone.
- **Sound is a window.** In a quiet position several moves are often within 3 win% of the best, so the arms are paired by position. A strong player's own move clears the window about two times in three.
- **Mostly blitz.** Most index games are blitz; his games are rapid.
- **The same month.** The held-out strong positions come from the same month as the index, so the same players can be on both sides; their result describes and decides nothing.
- **A coverage count ran first.** Before the rule was fixed, a partial index from the first games of the stream showed that his positions' nearest matches were far, and the index was grown to 400,000 games before any move was scored.
- **Added after the run.** The grouping by the nearest match's distance, the index-size curve and the two positions were looked at once the rule had been applied. They describe; a rule built on them needs positions this study did not see.